

GPS Technical Brief: Appendices

October 1, 2026 · Malaika

This is the full text behind Appendices A–D. Each citation links to its source, and the bracketed number beside it, like [7], points to the Brief Notes at the end.

Appendix A. More Policies and Where Each One Stands

The body covers the 2018 law and the \$10 million demonstration. These are the other policies, in order.

Status: Study only Testing Pending In force / done Deadline missed / unenforced

YEAR	POLICY AND WHAT IT DOES	WHERE IT STANDS
2010	Sec. 219, Coast Guard Authorization Act of 2010 Told DHS to study whether a single domestic backup to GPS is needed. (GPS.gov) [7]	Study only
2016	Sec. 1618, FY2017 defense bill Had DOD, DOT, and DHS jointly study technology-neutral requirements to back up and complement GPS. (GPS.gov) [7]	Study only
2017–18	Sec. 1606, FY2018 defense bill Told the three agencies to plan a backup GPS demonstration and authorized \$10 million, which Congress appropriated in March 2018. (U.S. DOT, 2021) [9]	Done DOT ran the demonstration in March 2020.
2018	National Timing Resilience and Security Act (49 U.S.C. § 312) Told DOT to provide a backup timing system within two years, subject to appropriations. A 2022 law (P.L. 117-103) struck the words “land-based.” (49 U.S.C. § 312) [8]	Deadline missed Deadline passed in December 2020. No operating federal backup.
2020–21	Executive Order 13905 and Space Policy Directive-7 Set policy toward responsible use of GPS, multiple sources of timing, and stronger GPS cybersecurity. (GPS.gov [21]; Office of Space Commerce [22])	In force Still current national policy.
2021	DOT report to Congress A roadmap for the 2018 law. Its industry roundtables found that no single system meets all needs and that carriers may not pay a subscription for backup timing. (U.S. DOT, 2021) [9]	Report only Not an implementation plan.
2024–26	DOT complementary PNT testing \$7.2 million to nine vendors for field tests. Phase I finished in June 2025, with a report to DOT leadership in March 2026. (U.S. DOT, 2024 [19]; Volpe Center, 2026 [20])	Testing Not deployment.

YEAR	POLICY AND WHAT IT DOES	WHERE IT STANDS
2025–26	FCC inquiry and NextNav petition The FCC opened an inquiry in 2025 (GPS World, 2025) [23]. NextNav asked for a nationwide 15 MHz license in the 902–928 MHz band (FCC, 2024) [24]. A draft rule went to the White House budget office in March 2026, and in April 2026 a House committee adopted an amendment to block FCC funds for finalizing it. (Broadband Breakfast, 2026) [25]	Pending Not law.
2025	S.2277 and H.R. 4344 S.2277 would require a DOD report on GPS risks, including a terrestrial backup framework. H.R. 4344 would direct an Air Force low-orbit navigation pilot. [26]	Pending Bills, not law.
2026	House FY2026 defense spending bill Added funding for Resilient GPS satellites that the Space Force had not requested. (SpaceNews, 2026) [27]	House-passed Not law.
2026	FAA GNSS Interference Resource Guide and GAO-26-108439 The FAA guide says it is not regulatory (FAA, 2026) [4]. GAO found no real-time detection and unfinished risk assessments for 7 of 8 systems, and made 9 recommendations. (GAO, 2026) [12]	Open FAA agreed to all nine; all still open.
2025	ICAO Assembly and Israel's ITU pledge ICAO condemned Russia and North Korea (ICAO, 2025) [30]. Israel pledged to limit interference to imminent threats and 15 minutes per incident. (Inside GNSS, 2026) [32]	Unenforced No penalty; the pledge is voluntary.

Appendix B. More Technology Solutions Beyond Xona

Each entry is one way to keep working when GPS fails, in plain terms, with the caveat.

LOW-ORBIT SATELLITES

Iridium PNT chip

Delivers secure time and location from Iridium's existing satellites. On sale since July 2026, and Iridium says 150+ organizations have asked about it (Iridium, 2026). Iridium claims its signal is up to 1,000 times stronger than GPS (Iridium) [36].

CAVEAT These are company claims.

GROUND-BASED RADIO

NextNav

A ground network in the 902–928 MHz band for positioning and timing (FCC, 2024) [24].

CAVEAT Contested at the FCC. Railroads, utilities, and toll operators object (Broadband Breakfast, 2026) [25], and critics say it needs only about 5% of the spectrum it seeks (R Street, 2026) [29].

SIGNED SATELLITE SIGNALS

Galileo OSNMA (Europe)

Each message carries a cryptographic signature so receivers can reject fakes. Live since July 24, 2025 (European Commission, 2025) [37].

CAVEAT It is Europe's system, not GPS.

MAGNETIC NAVIGATION

SandboxAQ AQNav

Reads Earth's magnetic field like a map, with AI filtering out the aircraft's own magnetic noise. Stayed within a standard accuracy tolerance for 100% of 150+ flight hours (The Quantum Insider, 2025) [38].

CAVEAT That result is company-reported, and a later drone test disclosed no accuracy data (Runtime Wire, 2026) [39].

CAMERA-BASED NAVIGATION

Ukraine's drones

Onboard AI matches camera images against maps when jamming cuts off GPS (The Defense Post, 2026) [40].

CAVEAT Battlefield use, with few public details.

DETECTION SOFTWARE

Machine-learning detectors

About 96% detection on drones in the study in the body (Gormus et al., 2026) [18]. In a driving simulation, a small deliberate shift in a spoofed position cut one detector's accuracy from 99.9% to 20.4% (PMC study) [41], and spoofing has also worked against AI-based drone navigation (arXiv, 2025) [42].

CAVEAT Simulations and controlled tests only. I found no field evidence of AI-enhanced spoofing.

WHAT DOT TESTED IN 2020

11 technologies

Two low-orbit satellite systems, two fiber-optic timing systems, one map-matching and inertial system, and six ground radio systems (U.S. DOT, 2021, p. 7) [9].

CAVEAT A demonstration only, and DOT found no single system meets every need.

Appendix C. Evidence on How Much Interference There Is

The key numbers come first, then the cases behind the headlines.

KEY NUMBERS

5,655 · 3,228 · 2,375

MOST SPOOFED AIRSPACE, JULY–AUG. 2024

Incidents over Nicosia, Tel Aviv, and Cairo (FAA, 2026, p. 49) [4].

1,100+

SPOOFED FLIGHTS PER DAY

Up from a few dozen in Feb. 2024 to more than 1,100 in Aug. 2024 (GPS World, 2024) [5].

+220%

GPS SIGNAL LOSS

Rise between 2021 and 2024, per IATA (IATA, 2025) [6].

~972

GULF SHIPS JAMMED PER DAY

Persian Gulf shipping, June 15–18, 2025 (Windward, 2025) [33].

\$1.4T · ~\$1B/day

ECONOMIC STAKES

Benefits since the 1980s, and the cost of a nationwide outage. The daily figure is a worst-case model; GPS outages have historically lasted under a day (RTI International, 2019) [3].

CASES

Flight 8243 (Dec. 2024). Jamming, then spoofing near Grozny. A missile struck the plane and killed 38. The interference was not the cause of the crash (FAA, 2026, p. 14) [4].

A pilot report, Amman. A crew judged a terrain alert false and did not start an escape maneuver, because their landing system and air traffic control agreed on their position (FAA, 2026) [4].

U.S. events. Interference near Denver and Dallas in 2022 (FAA, 2026, p. 10) [4], and a New Mexico air ambulance crash in May 2026 while military jamming was underway. The NTSB has not determined a cause (AVweb, 2026) [11].

Front Eagle (June 2025). The tanker collided with another tanker on June 17, during heavy Gulf jamming. Its operator blamed an “unspecified navigational error” (Windward, 2025) [33].

Baltic and Black Sea. Baltic interference has been almost daily since February 2022, and the Black Sea is described as a “gray zone” (Defense News, 2025 [34]; Spire, 2026 [35]).

Use with care: the Gulf jamming and the Front Eagle cause are not formally attributed. Counts from companies use different methods, so treat them as trends, not exact totals.

Appendix D. Likely Press Questions

Short answers the office can use if the press asks.

Q Is it safe to fly near conflict zones?

Pilots are trained to fall back on other navigation, and the FAA's guide walks them through it, but interference is rising and spoofing is hard to spot (FAA, 2026) [4].

Q Why don't we have a backup?

Congress ordered one in 2018 but made it subject to appropriations. DOT ran a \$10 million demonstration and is still testing (U.S. DOT, 2021) [9].

Q Is the FCC about to give one company spectrum?

A draft rule is at the White House budget office, and a House committee moved to block it. Neither is final (Broadband Breakfast, 2026) [25].

Q Did jamming cause the Azerbaijan Airlines crash?

No. A missile struck the plane, though the crew was flying without GPS (FAA, 2026, p. 14) [4].

Q Doesn't our ally do the same thing as Russia?

Israel is widely reported to spoof defensively (ABC News, 2024) [31] and has pledged limits (Inside GNSS, 2026) [32]. We should push all states, including partners, toward the same limits.

Brief Notes on Sources

Each note says what the source is good for and where to be careful. The numbers match the bracketed numbers in the appendices.

1. GPS.gov, “GPS and Telling Time.” Official source for GPS as a timing system for telecom, power, and finance.
2. Federal Communications Commission fact sheet (Mar. 2025). What happens when timing fails: carrier networks lose sync, 911 suffers.
3. RTI International for NIST, *Economic Benefits of GPS* (2019). The \$1.4 trillion and \$1 billion-a-day figures. The daily figure is a worst-case model.
4. FAA, *GNSS Interference Resource Guide v1.1* (Mar. 2026). The main U.S. government source on jamming and spoofing. Advisory, not regulatory.
5. GPS World, “GNSS spoofing threatens airline safety...” (Sept. 24, 2024). Spoofed-flight counts from SkAI Data Services and ZHAW. SkAI is a university spin-off that sells tracking.
6. IATA/EASA press release (June 18, 2025). The 220% rise in GPS signal-loss events, 2021–2024.
7. GPS.gov, “GPS Legislation.” Text summaries of the 2010, 2016, 2017, and 2018 backup laws.
8. 49 U.S.C. § 312. Current law, including the two-year deadline and the 2022 “land-based” change.
9. U.S. DOT, *NTRSA Roadmap to Implementation* (Jan. 2021). Primary source for the \$10M demonstration, the roundtables, and “not an implementation plan.”
10. DHS, *Resilient PNT fact sheet* (Jan. 2023). Says civilian GPS signals are low-power and unencrypted.
11. AVweb, “GAO Flags FAA Jamming, Spoofing Gaps” (Sept. 2026). Covers the GAO report and the New Mexico crash. Trade press.
12. U.S. GAO, *GAO-26-108439* (Sept. 2026). Primary source for the FAA detection and risk-assessment gaps.
13. Xona, “Positioning the world for progress.” The company’s mission. Backers appear only as logos.
14. Xona, *Series C announcement* (Mar. 2026). Company account of its \$170M raise, investors, and receiver testing.
15. GPS World, “Lockheed Martin invests in Xona...” (2022). Independent confirmation of Lockheed’s investment.
16. SRI, “Xona secures a \$92 million funding round.” Shows SRI licensed technology to Xona. SRI is not an investor.
17. Xona, *Pulsar page*. Xona’s specs and claims. Marketing, not independent verification.
18. Gormus et al., *Computers & Industrial Engineering* (2026). Controlled test of ML detection on drones. I saw the abstract and snippets only.
19. U.S. DOT, “Awards \$7 Million for Complementary PNT” (2024). The main federal testing effort so far.
20. Volpe Center, *Complementary PNT Action Plan update* (Apr. 2026). Phase I done June 2025, report delivered March 2026.
21. GPS.gov, “Resilience Through Responsible Use of PNT.” Index of federal actions, including Executive Order 13905.
22. Office of Space Commerce, *U.S. Space-Based PNT Policy*. Summary of SPD-7, the current national PNT policy.
23. GPS World, *NextNav proposal* (May 2025). Confirms the 2025 FCC inquiry and lays out NextNav’s case.
24. FCC Public Notice DA 24-776 (2024). Official description of NextNav’s request.
25. Broadband Breakfast, “House Appropriators Move to Block NextNav” (Apr. 2026). Amendment language, opponents, and the March 2026 draft rule. Committee action only.
26. Congress.gov, S.2277 and H.R. 4344. Pending bills.
27. SpaceNews, “Space Force may be done with R-GPS...” (2026). Congress adding Resilient GPS funding the Space Force did not request.
28. NextNav, *Law360 repost* (2024). NextNav’s “no cost to taxpayers” claim. Advocacy.
29. R Street Institute, “NextNav’s National Security Theater” (2026). A critic’s view. Opinion.
30. ICAO, *Assembly condemns GNSS interference* (Oct. 2025). The formal international finding. No penalty.
31. ABC News (Australia), *Israel GPS spoofing* (Sept. 2024). The defensive logic of Israeli spoofing and civilian spillover.
32. Inside GNSS, *Secure World Foundation summary* (2026). Source for Israel’s 15-minute ITU pledge. Trade-press summary.
33. Windward, “Middle East on the Precipice” (June 2025). Gulf jamming and the Front Eagle collision. Commercial data.
34. Defense News, *Baltic jamming* (July 2025). Baltic interference since 2022. Not fully opened.
35. Spire Global, *Black Sea report* (2026). Company blog. Not fully opened.
36. Iridium press release (July 2026) and *PNT platform page*. Product on sale now. Company claims.
37. European Commission, “How Galileo OSNMA helps counter spoofing” (Sept. 2025). Explains signed signals. Live July 24, 2025.
38. *The Quantum Insider on SandboxAQ* (July 2025). Flight-test results. Company-reported.
39. *Runtime Wire on SandboxAQ* (Sept. 2026). Independent caution on the drone test.
40. *The Defense Post*, *Ukraine vision navigation* (Jan. 2026). Example of camera-based navigation.
41. PMC, “Adversarial Evasion Attacks on SVM-Based GPS Spoofing Detection.” Simulation only.
42. arXiv, “GPS Spoofing Attacks on AI-based Navigation...” (2025). Preprint, not peer-reviewed.